Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	28-10-2026	Team ID	EN25CS3011021
Project Name	EV Market Performance and Insights	Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Tableau calculated fields and LOD expressions	Used to create measures such as CO2 saved and retained value, and to filter, aggregate and compare data by category, manufacturer, battery type and region.
2.	Database / Data Warehouse	Warehouse used to store and manage structured business data for analysis and reporting.	Excel / CSV file (EV market dataset)	Holds the structured data on sales, price, battery, range, manufacturer, country, ratings and CO2, and connects to Tableau without needing a database server.

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3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Desktop data source connection (Excel / Text file)	Loads the dataset straight into Tableau and can be refreshed whenever the source file is updated.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Desktop: Data Source tab, calculated fields and aggregations	Manages data types, relationships and calculations for KPIs such as total sales, average price, average range, retained value and CO2 saved.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Desktop: bar, line, box plot, bubble, scatter, dual-axis and stacked area charts	Creates the interactive visuals for category sales, manufacturer performance, battery technology, driving range, charging efficiency and regional CO2 impact.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Dashboards (Dashboard 1 and Dashboard 2), shared through Tableau Public	Gives executives and analysts interactive filters, legend highlighting and tooltips without coding, and lets stakeholders open the dashboards in a browser.